# ZS-F41
# A Conditional Odd Charge-Lattice Candidate from the Binary Pentagon Lift: USp(10), the $A_9/C_5$ Fold, the Bipartite Mediation-Parity Theorem, and the Negative Execution of the Sixth and Quaternionic Routes to M_UV

**Author:** Kenny Kang (Z-Spin Collaboration) **Date:** March 2026 **Theme / Paper code:** Foundations / **ZS-F41** — UV odd-sector axiom derivation; supplies the first conditional candidate for gate **ZS-F32 §8.5 B3-1**; addresses the pre-registered reopening trigger **F-F40.6**; registers the **Terminal Stopping Rule** for the B3 absolute-scale program **Version:** v1.1 (supersedes v1.0; change log in Version History)

**Verification: 31/31 PASS + 4/4 guards** (zs_f41_verify_v1_1.py) | **Zero Fitted Parameters** | Locked inputs: **A = 35/437**, **Q = 11**, (Z, X, Y) = (2, 3, 6), dim Z = 2 | **One structural condition: (H-SPIN).** Auxiliary coupling conditions, value audits only: **(H-BF₋)** (stage 1), **(H-QM)** (stage 2) | One firewalled external-anchored number: ν_now = 276.6 (ZS-A26), entering §§7–8 comparison only

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## §0. Abstract

ZS-F40 v1.2 closed the graded-determinant program for the absolute odd-sector susceptibility $\chi_-$ and left exactly one reopening gate, F-F40.6: *derive the odd gauge group and its charge lattice from the axioms* (gate B3-1 of ZS-F32 §8.5), *fix $\chi_-$ parameter-free, and land ρ_A in the observed window*. F40 Appendix D fixed the character of any successor: an **axiom-derivation paper**. This paper is that successor, and it reports all of its halves honestly: **the lattice half succeeds at the conditional-candidate tier; the value half fails decisively — twice, under two separately frozen pre-registrations — and the near-miss coefficient $b_-$ = 25/2 suggested by the first failure is proven unreachable in every derived frame. The B3 absolute-scale program is therefore terminated under a registered stopping rule inside this paper: no successor B3 paper exists or will be opened.**

The derivation transplants the corpus's own executed even-sector template — the ZS-M9 McKay chain ($Z_5 \to \hat{A}_4 \to SU(5) \to$ SM, zero free parameters), the ZS-U9 Trinity Braiding Theorem, and the ZS-S10 gauge bridge ($\kappa^2 = A/Q$) — to the seam-odd sector, under a single structural condition, **(H-SPIN)**: *the ZS-F23 seam parity β is the central {±1} of the Z-Spin SU(2), the 4π Z-Spin closure element.* Under (H-SPIN):

1. **Binary Pentagon Lift (Thm. F41.2, CONDITIONAL-CANDIDATE).** The seam-odd counterpart of the M9 pentagon stabilizer $Z_5$ is its binary preimage $\mathbf{Z_{10}}$ ($g^5 = -1$, check B1).
2. **Bipartite Mediation-Parity Theorem (Thm. F41.3, PROVEN).** The McKay graph of $Z_{10}$ is the 10-cycle $\hat{A}_9$; Z-Spin mediation (⊗**2**) is strictly parity-reversing; the graph is bipartite with color classes = (seam-even, seam-odd), and the seam-even class is verbatim the ZS-M9 $\hat{A}_4$ chain

(**Even-Sector Recovery**, checks B2–B4). A McKay cycle admits the seam grading iff it is even iff the binary lift is present — the graph-theoretic 4π statement.

3. **Quaternionic Seam Fold (Thm. F41.4, mathematics PROVEN).** The canonical conjugation involution (unique involution fixing $\rho_0$; unique spinorial fixed node = the seam sign character $\rho_5$) folds su(10) under $j^2 = -1$ into the type A II pair: fixed algebra **usp(10)** — we write **USp(10)** for the compact symplectic group on $\mathbb{C}^{10}$ (dim **55 = Q(Q−1)/2**; sp(5) in the mathematicians' convention; "Sp(10)" in v1.0 corpus shorthand) — with seam-odd module $\mathbf{p = \Lambda^2_0}$, dim **44 = 4Q**. The 2π alternative $j^2 = +1$ would give so(10), dim 45: discriminated by the axiom, not chosen (checks C1–C4).
4. **Charge lattice (Thm. F41.5, IMPORTED-PROVEN + conditional identification).** $H^2$ of the minimal resolution of $\mathbb{C}^2/Z_{10}$ is the $\mathbf{A_9}$ **root lattice** (discriminant $10 = |Z_{10}|$; McKay–GSV–Kronheimer), seam-split rank $5 \oplus 4$, folding onto $\mathbf{C_5}$ with $\rho_5$ on the long root (checks D1–D3); compactness and completeness imported from Harlow–Ooguri and Heidenreich et al. under $Z = \partial X$ read holographically.
5. **Stage-1 value audit (§7, CLOSED-NEGATIVE under (H-SPIN) $\wedge$ (H-BF$_-$)).** The Sixth Route — transmutation with derived content, the route F40's five-route audit gated rather than closed — is executed on a universe of **16 readings frozen before evaluation**: all 16 miss $\nu\_now = 276.6$, min $|\Delta\nu|$ = **37.60** (a factor $\geq 2\times10^{16}$ in $\rho\_\Lambda$).
6. **Stage-2 value audit and the 25/2 No-Go (§8, new in v1.1).** The stage-1 firewalled observation $b\_req = 12.4976 \approx 25/2$ is confronted head-on under a second frozen registration. The *derivable* part: the hyperkähler geometry of the Kronheimer construction motivates the quaternionic multiplet content, giving $\mathbf{b = 2h^\vee(C_5) = 12}$ exactly (condition (H-QM); NSVZ-protected; N = 4 consistency $b \to 0$; check F1) — and it **misses** ($\nu = 288.05$, +4.05 above the window; the entire Witten-admissible family $b \leq 12$ gives $\nu \geq 288.05$: family CLOSED-NEGATIVE, checks F2–F3). The *underivable* part: **Theorem F41.8 (25/2 No-Go)** — $b_- = 25/2$ is unreachable in every derived frame: (i) the Weyl frame requires $\Sigma T\_f = 57/4 \notin \frac{1}{2}\mathbb{Z}$ (impossible without unforced scalars, check F5); (ii) the quaternionic frame is capped at 12 with integer Witten-admissible steps (the half-step is Witten-forbidden, check F4); (iii) the APS boundary correction carries the opposite sign, $-(\eta+h)/2 \leq -\frac{1}{2}$ (toward 23/2, $\nu = 300.56$, miss). Even granted, $\nu(25/2) = 276.545$ falls below the frozen window edge (check F6). O1 is therefore permanently OBSERVATION.

**Verdict (§11).** Gate B3-1: OPEN → **CONDITIONAL-CANDIDATE (derived under (H-SPIN))** — the first canonical candidate (USp(10), 44, $A_9/C_5$) supplied to the gate. Both transmutation stages: CLOSED-NEGATIVE. F-F40.6 does not fire; B3 is not closed; the F40 terminus stands, sharpened into the **Hierarchy-Generator Obstruction** and terminated by the **Stopping Rule**: *no successor B3 paper is opened unless the corpus first derives, as a theorem, either the odd coupling boundary value $\alpha_- \neq \kappa^2$ or a non-transmutation hierarchy generator (named candidate: the Quillen-curvature susceptibility). Numerical proximity, alternative determinant choices, or new reading universes do not reopen B3.* The fractional face $\Omega\_\Lambda = 83/121$ is a separate corpus result and is untouched.

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## Epistemic Status Legend

| Tag | Meaning |
|---|---|
| **PROVEN** | Exact mathematics, machine-verified in zs_f41_verify_v1_1.py, or imported from the refereed literature with citation. |
| **IMPORTED-PROVEN** | External theorem consumed verbatim; the Z-Spin identification that makes it applicable carries its own (weaker) tag. |
| **DERIVED** | Follows from locked corpus inputs with no new assumption. |
| **DERIVED-CONDITIONAL** | Follows given one or more *named* conditions, each with a stated promotion path. |
| **CONDITIONAL-CANDIDATE** | A canonical candidate derived under named conditions — the first object of its kind supplied to a gate — without claim of unconditional existence or of uniqueness beyond template fidelity. |
| **HYPOTHESIS-strong / -weak** | Registered conjecture with / without a pre-registered anti-numerology audit. |
| **OBSERVATION (FIREWALL)** | Numerical proximity recorded for tracking; zero evidential weight; may never be consumed by a derivation. |
| **OPEN / OPEN-TERMINAL-final** | Not decidable with current corpus tools / same, with the stopping rule engaged. |
| **CLOSED-NEGATIVE** | Pre-registered test executed and failed; the claim is retired under the stated conditions. |
| **NOT-DERIVABLE** | A target value proven unreachable within all registered derivation frames. |
| **NON-CLAIM** | Explicit disclaimer of an adjacent claim this paper does not make. |

**Notation.** We write **USp(10)** for the compact symplectic group acting on $\mathbb{C}^{10}$; its Lie algebra is sp(5) in the mathematicians' convention (usp(10) as a real form) and has dimension 55. In internal corpus shorthand (and in v1.0 of this paper) this group was denoted Sp(10). "Z-Spin mediation" denotes the dynamical action of the Z-Spin operator (tensoring by the fundamental doublet **2**); "Z-sector" denotes the 2-dimensional geometric stage. The seam $\mathbb{Z}_2$ grading is β (ZS-F23).

# §1. Introduction: the mandate, and what v1.1 adds

ZS-F40 v1.2 is a terminus. Its two gates and its five-route audit closed every graded-determinant path to the absolute normalization of $\chi_- = e_-^2/(4\pi^2 Z_-)$, whose dimensionless factor **36A/Q = 1260/4807** is already DERIVED (ZS-F35; re-verified as check E2). The Charge-Unit Obstruction (ZS-F33.8, PROVEN) explains why: flux integrality fixes numbers, never units. F40 §12 registered a single reopening trigger, F-F40.6 (derive the odd group and lattice; fix $\chi_-$ parameter-free; land ρ_Λ), and Appendix D fixed the successor's character as an axiom-derivation paper. F-F40.10 added that exhibiting a genuine sixth route is itself an advance; the deep-exploration record (Appendix D) located that route in ZS-F32 Table A3: transmutation with $b_-$, "OPEN (not derivable — needs odd matter content)" — gated on B3-1, not closed.

Version 1.0 of this paper executed the mandate and reported an honest split verdict: lattice half conditional-positive, value half negative on a frozen 16-reading universe. Version 1.1 integrates the post-review program in full, replacing any successor paper:

(a) the editorial corrections of the review (title tier, USp(10) notation, condition accounting, B3-1 tier language, stopping rule); (b) **stage 2**: the one derivable coefficient suggested by the review's closure analysis — the quaternionic-multiplet value $b = 2h^\vee(C_5) = 12$ — registered, evaluated blind, and closed negative at the family level (§8.1–8.2); (c) **Theorem F41.8 (25/2 No-Go)**: the proof that the seductive near-miss b_req ≈ 25/2 is unreachable in every derived frame, converting the strongest remaining temptation into a closed door (§8.3); (d) the **Terminal Stopping Rule** (§11), by which the B3 absolute-scale program ends inside this paper.

The ordering discipline is unchanged: §2 freezes everything — including the stage-2 universe, with full disclosure that it was registered after stage 1's evaluation and is therefore held to a theorem-or-nothing standard — before any value in §§7–8 is computed.

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# §2. Locked inputs, named conditions, and the two frozen evaluation universes

## 2.1 Locked inputs (consumed verbatim; checks A1–A5)

- Geometric impedance **A = 35/437**; register **Q = 11 = 2 + 3 + 6**; (Z, X, Y) = (2, 3, 6); $\Sigma d_i^2 = 49$.
- i-tetration locked dynamics: $z^* = 0.4382829367 + 0.3605924719\, i$ (residual $< 10^{-50}$), $\lambda^* = (i\pi/2)z^*$, μ = 0.1148346250, ω = 2.2592495540, **$\omega^2/2$ = 2.5521042734**.
- Z-Spin-mediated coupling **$\kappa^2$ = A/Q = 35/4807** (Block Fiedler Mediation Theorem, ZS-M6 T1, PROVEN); the ZS-M8 boundary law $\alpha\_EM = \kappa^2 + c_4\kappa^4$.
- Odd-block structure $\chi_- = (\dim Y)^2(A/Q)C\_norm\, M\_UV^4 = (1260/4807)C\_norm\, M\_UV^4$, C_norm = 1 (ZS-F35/F40.7); $\rho\_\Lambda = \tfrac{1}{2}\chi_-\omega^2$ (ZS-F32).
- Seam parity β (ZS-F23); 4π Z-Spin closure (ZS-A7); geometrized outer automorphism (ZS-F34.BIV); compact 3-form promotion (ZS-F32.24); bedrock **Z = ∂X** (ZS-Q12 / O-Q16.12).

- Firewalled external-anchored comparison number, **§§7–8 only**: $\nu\_now \equiv \ln(\bar{M}\_P^4/\rho\_\Lambda,obs)$ = **276.6** (ZS-A26, reduced-$\bar{M}$_P), window **[276.6, 284.0]** (ZS-F32.27 convention band; observational uncertainty $\delta\nu \approx \pm 0.05$).

**Non-expansion rule (guard G1).** No constant outside this list enters any theorem.

## 2.2 The structural condition

> **(H-SPIN).** The ZS-F23 seam parity β is the central {±1} of the Z-Spin SU(2); equivalently, β is the 4π Z-Spin closure element: seam-odd states are precisely those on which a 2π Z-Spin rotation acts as −1.

Status: named condition, not proven. Corpus support: ZS-A7 (axiomatic 4π closure on the Z-anchor spinor), ZS-F30 (seam swap Ŝ exchanging the two su(2) factors), ZS-F23 (β the unique seam $\mathbb{Z}_2$). **Registered promotion obligation (F-F41.3):** prove the naturality square $\beta \Rightarrow g^5 = -1 \in Z_{10}$ and $\theta_1$-odd spin structure $\Rightarrow \rho_5$ (the seam sign character) commute — i.e. that the F23 parity, the F36/F34 odd theta characteristic, and the F41 fixed node are one $\mathbb{Z}_2$. Success strengthens the lattice; it does not close the value (NC-F41.5).

## 2.3 The auxiliary coupling conditions (value audits only)

> **(H-BF₋)** [stage 1]. The odd 3-form coupling at the seam boundary $\bar{M}$_P is Z-Spin-mediated in the Block Fiedler sense: $\alpha_-(\bar{M}\_P) = \kappa^2$ at leading order — the odd analogue of ZS-M8. Honest flag: the weakest pillar; its alternative normalization $g_-^{-2} = \kappa^2$ is carried in the stage-1 universe.
>
> **(H-QM)** [stage 2, new in v1.1]. The seam-odd effective content is the quaternionic multiplet forced by the hyperkähler geometry of the Kronheimer construction under $j^2 = -1$: gauge field + two adjoint Weyl fermions + one adjoint complex scalar, the unique content whose coupling is protected by the quaternionic structure (NSVZ-exact; $b = 2h^\vee$).

Accounting, per review: **one structural condition (H-SPIN); two auxiliary coupling conditions, each confined to its own value audit.** No theorem of §§3–6 uses (H-BF₋) or (H-QM).

## 2.4 Frozen universe, stage 1 (16 readings; guard G3)

Chains {$Z_{10}$-fold USp(10) + 1 Weyl **44**, b = 58/3 (primary); USp(10) quenched, b = 22; $2D_5 \to SO(14)$ quenched, b = 44; $2I \to E_8$ quenched, b = 110} × couplings {$\alpha_- = \kappa^2$ (primary); $g_-^{-2} = \kappa^2$} × powers {4 (ZS-F35); 8 (ZS-A28)}. Formulas frozen: $S = 8\pi^2/(b\,g_-^{-2})$, $\nu\_pred = pS + 0.40204$. Verdict rule frozen: hit iff $\nu\_pred \in [276.6, 284.0]$; no post-hoc reading (F-F41.7).

## 2.5 Frozen universe, stage 2 (2 readings; guard G4) — disclosure

Stage 2 was formulated **after** the stage-1 evaluation, in response to its firewalled observation O1 (b_req ≈ 25/2). It is therefore held to the strictest standard: **only coefficients that arrive as theorems from target-blind ingredients are evaluated as readings.** Exactly one such coefficient exists — the quaternionic-multiplet value $b = 2h^\vee(C_5) = 12$ under (H-QM) — giving the stage-2 universe {b = 12} × {$\alpha_- = \kappa^2$} × {p = 4, 8} = **2 readings**. Non-integer candidates (25/2; the APS-signed 23/2) enter §8.3 **only as No-Go arithmetic, never as claim readings**. The same frozen verdict rule and window apply.

## §3. The even-sector template audit (Theorem F41.1)

**Theorem F41.1 (Template Existence; DERIVED, corpus reading).** The corpus has already executed, once and in full, "derive the gauge group and the charge lattice from the axioms" — in the seam-even sector: (a) **group** — ZS-M9: pentagon stabilizer $Z_5 \subset SU(2)$ → McKay $\hat{A}_4$ → su(5) → Georgi–Glashow → SM, zero free parameters at every arrow (Table-2 assignment = Gap G2, HYPOTHESIS-strong, inherited and disclosed); (b) **lattice** — ZS-U9 Trinity: the full hypercharge spectrum {+1/6, +2/3, −1/3, −1/2, −1, 0} and Q_e = −1 as outputs, the five anomaly conditions emerging rather than imposed; (c) **coupling unit** — ZS-S10 + ZS-M8: $\alpha\_EM = \kappa^2 + c_4\kappa^4$, $1/\alpha = 137.0359$ (1.07 ppm). Gate B3-1 is therefore the seam-odd re-execution of an executed pattern, with dictionary: $Z_5$ → binary lift; $\hat{A}_4$ → even-cycle counterpart; Cartan lattice → Kronheimer lattice; $\alpha = \kappa^2$ → (H-BF$_-$).

## §4. The Binary Pentagon Lift and the Bipartite Mediation-Parity Theorem

### 4.1 The lift

**Theorem F41.2 (Binary Pentagon Lift; CONDITIONAL-CANDIDATE under (H-SPIN)).** The seam-odd counterpart of the M9 input $Z_5$ is its binary preimage under SU(2) → SO(3): $\mathbf{Z_{10} = \langle g\rangle, g = diag(\zeta, \bar{\zeta}), \zeta = e^{i\pi/5}, g^{10} = 1, g^5 = -1}$ (check B1). *Proof sketch:* by (H-SPIN) the seam-odd sector is acted on by $-1 \in SU(2)$; $Z_5$ (odd order) intersects {±1} trivially and sees only the seam-even story; the unique template-faithful enlargement containing −1 is $\langle Z_5, -1\rangle = Z_{10}$. The lifts of the larger pentagon stabilizers ($2D_5$, 2I) are carried as disclosed alternatives in §2.4.

### 4.2 The parity theorem

**Theorem F41.3 (Bipartite Mediation-Parity; PROVEN, checks B2–B4).** (i) The McKay graph of $Z_{10}$ is the 10-cycle $\hat{A}_9$: $\mathbf{2} \otimes \rho\_k = \rho\_\{k-1\} \oplus \rho\_\{k+1\}$. (ii) Z-Spin mediation is strictly parity-reversing: every McKay edge joins a vectorial node (k even, β = +1) to a spinorial node (k odd, β = −1); the graph is bipartite with color classes equal to the seam grading. (iii) (**Even-Sector Recovery**) The vectorial class is Rep($Z_5$) via $\zeta^2 = \omega_5$, its McKay graph the 5-cycle $\hat{A}_4$ — ZS-M9 reproduced verbatim inside the odd construction.

**Corollary (graph-theoretic 4π).** A McKay cycle admits the seam grading iff it is even iff $-1 \in \Gamma$ iff the binary lift is present. $\hat{A}_4$ (odd cycle) admits none — which is *why* the odd sector required the lift. The 4π Z-Spin closure, the bipartiteness of mediation, and the existence of the odd sector are one statement in three languages.

## §5. The Quaternionic Seam Fold: $G_-$ = USp(10) with matter 44

### 5.1 The involution is canonical

On $\hat{A}_9$ the conjugation ρ_k ↔ ρ-k is (a) the canonical duality of the character ring, (b) the geometrization supplied by ZS-F34.BIV, and (c) **the unique nontrivial graph involution fixing $\rho_0$** (check B6). Its fixed nodes are $\rho_0$ and $\rho_5$; $\rho_5$ is spinorial with $\rho_5(g) = -1$, $\rho_5(g^2) = +1$ — **the seam sign character**, the unique spinorial fixed point (check B5).

### 5.2 The 4π discriminator and the fold

**Theorem F41.4 (Quaternionic Seam Fold; mathematics PROVEN, checks C1–C4; physical assignment CONDITIONAL-CANDIDATE under (H-SPIN)).** The $4\pi$ closure is $\mathbf{j^2 = -1}$: the Z-Spin doublet is pseudo-real, realized by $J = [[0, I_5], [-I_5, 0]]$, $J^2 = -I$. The involution $\theta\_J(X) = J\bar{X}J^{-1}$ of su(10) has fixed subalgebra **usp(10)** (the Lie algebra of **USp(10)**), dim **55 = Q(Q−1)/2**, with symplectic witness $X^TJ + JX = 0$, and anti-invariant module $\mathbf{p = \Lambda^2{}_0(10)}$, dim **44 = 4Q** — the type A II symmetric pair, moduli of quaternionic structures. Had the closure been $2\pi$ ($j^2 = +1$), the fixed subalgebra would be so(10), dim 45 (check C3): **the fold is discriminated by the axiom, not chosen.** Physical assignment: $\mathbf{G_- = USp(10)}$; p = **44** is the seam-odd matter module; $\rho_5$ folds onto the long root of $C_5$ (§6). The exact relations dim $G_-$ = Q(Q−1)/2, dim p = 4Q are OBSERVATION (FIREWALL): outputs, consumed by nothing.

### 5.3 Stage-1 content, anomaly safety, and $b_-$

**Theorem F41.6 (content arithmetic; PROVEN given the assignment; assignment CONDITIONAL-CANDIDATE).** With one Weyl multiplet in $p = \Lambda^2{}_0$ (minimal dynamical realization of a nonempty seam-odd sector; $\Lambda^2{}_0$ real, hence Majorana-compatible, no cubic anomaly): machine-verified indices T(fund) = ½, T(adj) = 6, $T(\Lambda^2{}_0) = 4$ (check C5); Witten global anomaly ($\pi_4(USp(10)) = \mathbb{Z}_2$): safe, 2T = 8 even (check C6); $\mathbf{b_- = (11/3)T(adj) - (2/3)T(\Lambda^2{}_0) = (11\cdot 6 - 2\cdot 4)/3 = 58/3}$; quenched b = 22; disclosed alternatives SO(14) b = 44, $E_8$ b = 110.

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# §6. The charge lattice

**Theorem F41.5 (Charge Lattice; IMPORTED-PROVEN mathematics + CONDITIONAL-CANDIDATE identification).** (a) For $\Gamma = Z_{10} \subset SU(2)$, $H^2$ of the minimal resolution of $\mathbb{C}^2/\Gamma$ is the **$A_9$ root lattice** (McKay [1]; Gonzalez-Sprinberg–Verdier [2]; Kronheimer [3,4]), discriminant det $C(A_9)$ = **10 = |$Z_{10}$|** (check D1); the hyperkähler nature of Kronheimer's construction is the geometric twin of the §5 fold — the same $j^2 = -1$ governs both. (b) The seam involution $\alpha\_k \leftrightarrow \alpha\_\{10-k\}$ splits the lattice rank 5 ⊕ rank 4 (check D2); the orbit-folded Cartan is **$C_5$** (det 2), the fixed node — image of $\rho_5$ — on the **long root** (check D3); the F32.24 odd flux lattice is identified with this structure (the ALE-carrier identification being the conditional step). (c) Compactness of $G_-$ and completeness of the charge spectrum are theorems in holographic quantum gravity — Harlow–Ooguri [5,6], extended to arbitrary compact groups by Heidenreich et al. [7] — hence theorem-conditional on the bedrock $\mathbf{Z = \partial X}$ read holographically; if that reading is ever closed negatively, this leg reverts to the F32.24 assumption tier (F-F41.8-gate) while (a)–(b) stand on finite mathematics.

**Status of B3-1 after §§4–6:** OPEN → **CONDITIONAL-CANDIDATE (derived under (H-SPIN))** — the first canonical candidate (USp(10), 44, $A_9/C_5$) supplied to the gate, where before there was a free choice. This is the precise sense, no stronger, in which F-F40.6(i) is addressed.

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## §7. Stage 1: the Sixth Route, executed blind

With derived content, transmutation is parameter-free given a coupling boundary: $S = 8\pi^2/(b_- g_-^2)$, $M\_UV = \bar{M}\_P\, e^{-S}$, $\nu\_pred = pS +$ **0.40204** (check E1). The frozen 16-reading audit (checks E3–E5):

| # | Chain | Coupling | p | S | ν_pred | \|Δν\| |
|---|---|---|---|---|---|---|
| 1 | **USp(10) + 1 Weyl 44 (primary)** | **$\alpha_- = \kappa^2$ (primary)** | **4** | **44.635** | **178.94** | **97.66** |
| 2 | USp(10) + 1 Weyl 44 | $\alpha_- = \kappa^2$ | 8 | 44.635 | 357.49 | 80.89 |
| 3 | USp(10) + 1 Weyl 44 | $g_-^2 = \kappa^2$ | 4 | 560.90 | 2244.0 | 1967.4 |
| 4 | USp(10) + 1 Weyl 44 | $g_-^2 = \kappa^2$ | 8 | 560.90 | 4487.6 | 4211.0 |
| 5 | USp(10) quenched | $\alpha_- = \kappa^2$ | 4 | 39.225 | 157.30 | 119.30 |
| 6 | **USp(10) quenched** | **$\alpha_- = \kappa^2$** | **8** | **39.225** | **314.20** | **37.60** |
| 7 | USp(10) quenched | $g_-^2 = \kappa^2$ | 4 | 492.92 | 1972.1 | 1695.5 |
| 8 | USp(10) quenched | $g_-^2 = \kappa^2$ | 8 | 492.92 | 3943.7 | 3667.1 |
| 9 | SO(14) quenched ($2D_5$) | $\alpha_- = \kappa^2$ | 4 | 19.613 | 78.85 | 197.75 |
| 10 | SO(14) quenched ($2D_5$) | $\alpha_- = \kappa^2$ | 8 | 19.613 | 157.30 | 119.30 |
| 11 | SO(14) quenched ($2D_5$) | $g_-^2 = \kappa^2$ | 4 | 246.46 | 986.23 | 709.6 |
| 12 | SO(14) quenched ($2D_5$) | $g_-^2 = \kappa^2$ | 8 | 246.46 | 1972.1 | 1695.5 |
| 13 | $E_8$ quenched (2I) | $\alpha_- = \kappa^2$ | 4 | 7.845 | 31.78 | 244.8 |
| 14 | $E_8$ quenched (2I) | $\alpha_- = \kappa^2$ | 8 | 7.845 | 63.16 | 213.4 |
| 15 | $E_8$ quenched (2I) | $g_-^2 = \kappa^2$ | 4 | 98.583 | 394.74 | 118.1 |
| 16 | $E_8$ quenched (2I) | $g_-^2 = \kappa^2$ | 8 | 98.583 | 789.07 | 512.5 |

**Result: all 16 miss.** min $|\Delta\nu|$ = **37.60** (reading 6: undershoots $\rho\_\Lambda$ by $e^{37.6} \approx 2.1\times10^{16}$); the primary reading misses the other side by $e^{97.7}$. The misses span both signs and three decades in $|\Delta\nu|$: robust against every disclosed convention, hence no look-elsewhere Monte Carlo is required for the negative.
**The Sixth Route is CLOSED-NEGATIVE under (H-SPIN) ∧ (H-BF₋).**

# §8. Stage 2 (new in v1.1): the quaternionic route and the 25/2 No-Go

## 8.1 The one derivable coefficient

The review's closure analysis proposed $b_- = 25/2 = 2h^\vee(C_5) + ½$. Stage 2 separates the derivable half from the underivable half. The derivable half: under (H-SPIN), the charge lattice lives on a **hyperkähler** space (Kronheimer, §6a) — the geometry whose defining structure is the same $j^2 = -1$ that selected the fold. The gauge content compatible with (i.e. protected by) the quaternionic structure is the quaternionic multiplet — gauge field, two adjoint Weyl fermions, one adjoint complex scalar — whose coefficient is exact and scheme-independent by the NSVZ non-renormalization structure [11]:

$$\mathbf{b = (11\cdot 6 - 2\cdot(6+6) - 1\cdot 6)/3 = (66 - 24 - 6)/3 = 12 = 2h^\vee(C_5)} \text{ (check F1),}$$

with the maximal-symmetry consistency check that adding the adjoint hypermultiplet gives $b = 12 - 12 = 0$. This is condition **(H-QM)**: registered, target-blind in its ingredients ($h^\vee(C_5) = 6$ was machine-verified in v1.0 before any stage-2 number existed).

## 8.2 Execution and family closure

Stage-2 readings (frozen, §2.5): ($b = 12$, $\alpha_- = \kappa^2$, $p = 4$): $S = 71.913$, $\boldsymbol{\nu = 288.05}$ — misses the window by **+4.05**; ($b = 12$, $p = 8$): $\nu = 575.70$ — misses (check F2). Moreover, every Witten-admissible quaternionic matter addition lowers b by an **integer** (fundamental hyper $\Delta b = -1$; $\Lambda^2_0$ hyper $\Delta b = -8$; adjoint hyper $\Delta b = -12$; the half-step — a fundamental half-hypermultiplet, legal in principle since **10** is pseudo-real — is excluded by the Witten global anomaly, $2T = 1$ odd; check F4). Since $\nu_4(b)$ is decreasing in b, the entire admissible family $b \in \{12, 11, \dots\}$ satisfies $\nu_4 \geq 288.05 > 284.0$ (check F3, $b = 1\dots12$ enumerated):

**the quaternionic route is CLOSED-NEGATIVE at the family level under (H-SPIN) $\wedge$ (H-QM) $\wedge$ ($\boldsymbol{\alpha_- = \kappa^2}$).**

The target b_req = 12.4976 sits in the gap (12, 58/3) between the two derived frames — above the quaternionic ceiling, below the Weyl-frame floor — a gap no derived content occupies.

## 8.3 Theorem F41.8 (the 25/2 No-Go)

**Theorem F41.8 (25/2 No-Go; DERIVED-CONDITIONAL within the registered frames; checks F4–F6).** The coefficient $b_- = 25/2$ is unreachable in every derivation frame this paper registers:

(i) **Weyl frame (stage 1).** $(66 - 2\Sigma T\_f)/3 = 25/2$ requires $\Sigma T\_f = 57/4 \notin ½\mathbb{Z}$, while every fermionic index of USp(10) lies in $½\mathbb{Z}$: **no fermionic content whatsoever realizes 25/2** (check F5). Scalar-assisted solutions exist arithmetically (e.g. one Weyl **44** plus $\Sigma T\_s = 41/2$ of scalars) but are forced by nothing in the derivation — selecting one is precisely the post-hoc move F-F41.7 voids.

(ii) **Quaternionic frame (stage 2).** The coefficient is capped at $2h^\vee = 12$ and descends only in Witten-admissible integer steps (§8.2): matter strictly *lowers* b, and the sole half-step is anomaly-forbidden. $25/2 > 12$ is unreachable from within the frame (check F4).

(iii) **Boundary corrections.** The only structure that could exceed the ceiling is a boundary/defect term, and the APS index correction carries the **opposite sign**, $-(\eta + h)/2 \leq -\frac{1}{2}$ for the seam-fixed spinorial node $\rho_5$ ($h \geq 1$) [10] — pointing to $12 - \frac{1}{2} = 23/2$ ($\nu = 300.56$, miss), not $12 + \frac{1}{2}$. The review's remaining sub-route — a Lefschetz-weighted "$\frac{1}{2} \cdot \mathrm{Tr}(\sigma|H^2) = \frac{1}{2}$" — has no connecting theorem and is refused as reverse-engineered.

Even if 25/2 were granted by fiat, $\nu(25/2) =$ **276.545** falls 0.054 *below* the frozen window edge 276.6 (check F6). That this residue lies within ~1σ of the observational uncertainty of ν_now (±0.05) is recorded as FIREWALL O3 and nothing more: the frozen rule governs, and a value with no derivation has no claim to adjudicate. **Consequence: the stage-1 observation O1 (b_req ≈ 25/2) is permanently OBSERVATION — proven never promotable within the registered frames.**

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## §9. Firewalled observations (non-evidence)

- **O1.** b_req($\alpha_- = \kappa^2$, $p = 4$) = 12.4976 ≈ 25/2 (0.02%). Per Theorem F41.8, unreachable in every derived frame; permanently quarantined.
- **O2.** $S^*/2\pi = 10.9896$, within 0.10% of $Q = 11$ — the known ZS-A31/A32 modular-depth proximity (p_single = 0.50%), consumed as history, re-claiming nothing.
- **O3.** $\nu(25/2) = 276.545$ vs frozen edge 276.6: gap 0.054, within ~1σ of $\delta\nu\_{obs} \approx 0.05$. Recorded because honesty requires it; evidential weight zero (no derivation exists to carry it).

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## §10. The Planck-unit organizing box (NON-CLAIM)

The five Planck quantities organize the F40 → F41 arc, with one correction to the founding intuition: $E\_P = m\_P c^2$, so the independent set is {ℓ_P, t_P, m_P, T_P, **q_P**}, and **$q\_P = \sqrt{4\pi\varepsilon_0\hbar c}$ is the unique Planck unit free of G**. The five routes F40 closed were all metric-side instruments; the surviving unknown $e_-^2$ is charge-type; and $q\_P^2 = 2h/Z_0$ — charge² = action/impedance — is the same structural sentence as $\chi_- = e_-^2/(4\pi^2 Z_-)$, with A the geometric impedance. The even sector closed its dimensionless version ($\alpha = \kappa^2 + c_4\kappa^4$, ZS-M8); the odd 3-form charge carries mass² dimension, which is why the unit problem is real, why $\bar{M}\_P$ must enter, and why in this paper the lattice bridge could stand while both value bridges fell. Organizing observation only; consumed by nothing (NC-F41.3).

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## §11. Terminal verdict, closure ledger, and the Stopping Rule

| Item | Before F41 | After F41 v1.1 |
|---|---|---|
| B3 fraction Ω_Λ = 83/121 | separate face, DERIVED-CONDITIONAL | **untouched** (distinct from the absolute scale) |
| B3 determinant/clock program | TERMINAL (F40) | unchanged |

| Item | Before F41 | After F41 v1.1 |
|---|---|---|
| Gate B3-1 (odd group + lattice) | OPEN | **CONDITIONAL-CANDIDATE (derived under (H-SPIN))**: (USp(10), 44, $A_9/C_5$) |
| Sixth route (stage 1, 16 readings) | gated on B3-1 (F32 Table A3) | **CLOSED-NEGATIVE** under (H-SPIN) $\wedge$ (H-BF$_-$), min $\lvert\Delta\nu\rvert = 37.60$ |
| Quaternionic route (stage 2, family) | — | **CLOSED-NEGATIVE** under (H-SPIN) $\wedge$ (H-QM): $\nu \geq 288.05$ for all admissible $b \leq 12$ |
| $b_- = 25/2$ | firewalled near-miss (O1) | **NOT-DERIVABLE** (Thm. F41.8); O1 permanently OBSERVATION |
| B3-4 (blind $\rho\_\Lambda$ output) | untestable | executed twice; negative twice |
| F-F40.6 reopening trigger | armed | **not fired**; F40 terminus stands |
| $\chi_-$ absolute value | OPEN-TERMINAL | **OPEN-TERMINAL-final** (stopping rule engaged) |

**Terminal Stopping Rule (registered).** No successor B3 paper is opened unless the corpus first derives, **as a theorem from the axioms**, either (a) the odd coupling boundary value $\alpha_- \neq \kappa^2$, or (b) a non-transmutation hierarchy generator — the named candidate being the Quillen-curvature susceptibility $\chi_- = -\partial^2\_J \log Z_-|_0$ on the $A_9/C_5$ determinant line, which is registered here as a candidate and **not** pursued as a paper. Numerical proximity (including O1–O3), alternative determinant choices, new reading universes, or re-derivations of existing negatives do not reopen B3. Theorem first; paper second; otherwise silence.

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# §12. Falsification gates

- **F-F41.1** (mathematical collapse — immediate retraction): any error in §4's character theory voids Thms. F41.2–F41.3. Coverage: B1–B6.
- **F-F41.2** (fold collapse): fixed algebra ≠ usp(10) (dim ≠ 55) or odd module $\neq \Lambda^2_0$ (dim ≠ 44) retracts Thm. F41.4. Coverage: C1–C4.
- **F-F41.3** ((H-SPIN) status): if $\beta$ is proven *not* the central $\{\pm 1\}$, every conditional item reverts to OPEN. Promotion obligation: the §2.2 naturality square from ZS-F30 + ZS-A7.
- **F-F41.4** (template collapse): terminal failure of ZS-M9's Gap G2 removes the §3 anchor; §§4–6 mathematics survives.
- **F-F41.5** (content collapse): errors in $(T(\mathrm{adj}), T(\Lambda^2_0)) = (6, 4)$ or Witten parity retract $b_- = 58/3$. Coverage: C5–C6.

- **F-F41.6** (stage-1 value gate — fired NEGATIVE): re-evaluated only per the Stopping Rule.
- **F-F41.7** (anti-numerology): any post-hoc modification of either frozen universe to approach $\nu$ = 276.6 is void ab initio (inherits F-F32.27, F-F40.8). O1–O3 may never be consumed.
- **F-F41.8-gate** (holography reversion): negative closure of $Z = \partial X$ as holography reverts Thm. F41.5(c) to the F32.24 assumption tier; F41.5(a)–(b) unaffected.
- **F-F41.9** (stage-2 gates — fired NEGATIVE): if $b = 2h^\vee(C_5) = 12$ or the Witten step-parity arithmetic is in error (checks F1–F4), §8.2's family closure is re-examined; if the APS sign convention in Thm. F41.8(iii) is shown inapplicable to the seam geometry *and* a positive half-unit boundary term is derived as a theorem, Thm. F41.8 is re-opened — this is the sole mathematical route back to 25/2, and it must arrive as a theorem before any evaluation (Stopping Rule).
- **F-F41.10** (stopping-rule breach): any B3-scale successor document not preceded by a theorem satisfying §11(a) or §11(b) is void under corpus rules.

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# §13. Cross-version safety and observational non-collision

**Dependency audit.** No locked quantity is modified: $z^*$, $\mu$, $\omega$, A, Q, $\kappa^2$, 1260/4807 consumed verbatim (checks A1–A5, E2); all downstream consumers of ZS-M1/F32/F35 untouched. No F40 status is reversed; the route table is extended by two executed-negative entries (stage 1, stage 2), exactly the extension F-F40.10 licensed. M9's G2 and M8's $c_4$ inherited at their tags. v1.0 → v1.1 changes are additive and tier-lowering only (title, notation, B3-1 tier, condition accounting) plus the new §8 and Stopping Rule; no v1.0 PASS is weakened, and the v1.0 16-reading verdict is preserved bit-identical (checks E1–E5 unchanged).

**Observational non-collision.** Terminal claims are structural (a candidate group, module, lattice) and negative (two closed routes; one no-go): no surviving dimensionful prediction collides with Planck-2018 ΛCDM or Standard-Model couplings. Transmutation scales of failed readings are not asserted physical (NC-F41.2).

**Non-claims.** NC-F41.1: B3 is not closed; $\chi_-$ is not fixed. NC-F41.2: no IR spectrum or dark-sector phenomenology of USp(10). NC-F41.3: §10 is organizational. NC-F41.4: nothing in F40 is reversed. NC-F41.5: (H-SPIN) is not proven; its promotion would strengthen the lattice, not the value. NC-F41.6: the $2D_5/2I$ folds were not executed (quenched coefficients only). NC-F41.7 (new): (H-QM) is a registered condition, not a derivation of supersymmetry; no claim that the odd sector *is* an N = 2 theory is made — only that the quaternionic-protected coefficient is $2h^\vee$. NC-F41.8 (new): the Quillen-curvature candidate in §11 is named, not begun.

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# §14. Conclusion

F-F40.6 asked for three things. This paper supplies the first as a candidate it can defend — under one structural condition, the odd sector acquires a gauge group it did not choose (**USp(10)**, symplectic *because* $j^2 = -1$), matter it did not postulate (**44 = 4Q**), and a charge lattice it did not assume (**$A_9$ folding to $C_5$**, the seam character on the long root), with the mediation itself proven bipartite and the even chain recovered verbatim inside the odd one. And it closes the other two, twice over: sixteen frozen readings missed by no less than sixteen orders of magnitude; the one honestly derivable rescue coefficient, $2h^\vee(C_5) = 12$, missed with its entire admissible family; and the number the first failure dangled — 25/2, four parts in ten thousand from the sky's own requirement — is now proven unreachable from every direction the corpus can derive: fermions cannot reach it, the quaternionic ceiling sits below it, and the boundary term points the other way.

The vacuum-energy depth still lacks a hierarchy generator, and the corpus now knows precisely what shape that missing datum must have — and has bound itself, by the Stopping Rule, to say nothing more about it until a theorem speaks first. That is not defeat; it is the corpus keeping its own discipline at the exact moment the temptation to break it was strongest.

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## Acknowledgements & Code Availability

Verification suite: zs_f41_verify_v1_1.py (31 checks + 4 guards; exact rational arithmetic via fractions, 60-digit dynamics via mpmath, representation theory and folding via numpy; exits non-zero on any theorem-tier failure; no fail-open clause). The script prints the stage-1 16-reading audit, the stage-2 readings, the No-Go arithmetic, and the firewalled observations O1–O3 outside the PASS ledger.

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## Appendix A. Character data for the lift

$Z_{10} = \langle g\rangle$, $g = \mathrm{diag}(\zeta, \bar\zeta)$, $\zeta = e^{i\pi/5}$; $\rho_k(g) = \zeta^k$; $\beta(\rho_k) = (-1)^k$; $\mathbf{2}|_{Z_{10}} = \rho_1 \oplus \rho_9$. McKay: $\mathbf{2} \otimes \rho_k = \rho_{k-1} \oplus \rho_{k+1} \to \hat{A}_9$. Vectorial subring $\cong \mathrm{Rep}(Z_5)$ via $\zeta^2 = \omega_5$; its McKay graph $\hat{A}_4$ (ZS-M9). $\sigma$: $\rho_k \mapsto \rho_{-k}$; $\mathrm{Fix}(\sigma) = \{\rho_0, \rho_5\}$; $\rho_5(g) = -1$ the seam sign character; $\sigma$ unique among the 20 dihedral maps fixing $\rho_0$ (check B6).

## Appendix B. The fold in matrices

$J = [[0, I_5], [-I_5, 0]]$, $J^2 = -I$. $\theta_J(X) = J\bar{X}J^{-1}$: involutive automorphism (C1); eigenspaces $55 \oplus 44$ (C2); fixed elements $X^TJ + JX = 0$ (C4) — usp(10). Real-structure control $\theta_r(X) = \bar{X}$: so(10), dim 45 (C3). Index witnesses with $H = i\cdot\mathrm{diag}(I_5, -I_5)$: $\mathrm{Tr}_{\mathrm{fund}}\, H^2 = -10$, $\mathrm{Tr}_{\mathrm{adj}}\, H^2 = -120$ ($T(\mathrm{adj}) = 6$), $\mathrm{Tr}_{\Lambda^2}\, H^2 = -80$ with the J-singlet at 0 ($T(\Lambda^2_0) = 4$) (C5). Lattice fold: orbits $\{\alpha_1\alpha_9\}\{\alpha_2\alpha_8\}\{\alpha_3\alpha_7\}\{\alpha_4\alpha_6\}\{\alpha_5\}$ sum the $A_9$ Cartan to the $C_5$ Cartan with $a_{54} = -2$: the fixed node is the long root; det 10 → det 2 (D1–D3).

## Appendix C. Stage-1 audit, exact values

Prefactor $(\omega^2/2)(1260/4807) = 0.668951817$, ln = −0.40204324. $\kappa^2 = 35/4807$. $S(b; \alpha_- = \kappa^2) = 2\pi\cdot 4807/(35b)$; $S(b; g_-^{\,2} = \kappa^2) = 4\pi \times$ that. $b = 58/3$: $S = 2\pi\cdot 14421/2030 = 44.63538$; $b = 22$: 39.22503; $b = 44$: 19.61251; $b = 110$: 7.845006. $\min|\Delta\nu| = 37.6023$ (reading 6); primary $\Delta\nu = -97.6564$. Firewalled: $S^* = 69.04949$, $b_{\mathrm{req}} = 12.49757$, $S^*/2\pi = 10.98957$.

## Appendix D. Deep-exploration records (protocol Steps 0–5, condensed)

**Record 1 (v1.0).** Long list 7 (dropped: Q = 11 ↔ 11D brane import — no corpus identification, no hierarchy; direct non-ADE odd lattice — category error, "odd" is seam parity). Issues: gate logic

(PROVEN, textual); template existence (DERIVED-CONDITIONAL); transplant (executed in this paper); unit half (resolved negative by the frozen execution). Convergence $2 \to 0$.

**Record 2 (v1.1).** Long list 7 (dropped: immediate Quillen-route execution — cost of a full paper, registered as the named Stopping-Rule candidate instead; standalone no-go successor F42 — absorbed into §8 and §11 per the no-proliferation directive). Issues by leverage: I1′ stage-2 derivability — (a) $b = 2h^\vee = 12$ derivable under (H-QM), machine-checked, evaluated, family-negative; (b) the +½ underivable by all three proposed mechanisms (matter monotone-decreasing with the half-step Witten-forbidden; APS sign negative; Lefschetz-trace weighting refused as reverse-engineered); (c) 25/2 frame-exhaustive No-Go (Thm. F41.8). I2′ editorial soundness — five review items adopted. I3′ Stopping Rule — registered. I4′ (H-SPIN) naturality square — obligation recorded. Convergence: cycle 2 corrected two nodes (half-hyper/Witten interplay; the scalar-loophole wording from "impossible" to "unforced → F-F41.7-void"); cycle 3 zero changes — converged. External PROVEN imports supporting the No-Go: Witten [8], APS [10], NSVZ [11].

# Appendix E. Stage-2 exact values and No-Go arithmetic

b(quaternionic) = $(66 - 24 - 6)/3 = 12 = 2h^\vee(C_5)$; $N = 4$ limit $b = 0$. $\nu(12, 4) = 288.0523$ (window +4.05); $\nu(12, 8) = 575.7025$. Family: $\nu_4(b) = 8\pi\cdot 4807/(35b) + 0.40204$, decreasing; $\nu_4(11) = 314.2023$ (= stage-1 reading 6 by the arithmetic identity $8/22 = 4/11$); $\nu_4(b \le 12) \ge 288.05$. Weyl-frame requirement: $\Sigma T\_f = 57/4$ (denominator 4: outside $\tfrac{1}{2}\mathbb{Z}$); scalar loophole $\Sigma T\_s = 41/2$ (arithmetically valid, unforced). $\nu(25/2, 4) = 276.5462$ (gap to edge 0.0538); $\nu(23/2, 4) = 300.5588$. $b\_req - 25/2 = -0.00243$ (0.019%).

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## Version History

v1.0 (March 2026): Initial public release. Lattice derivation under (H-SPIN); stage-1 frozen 16-reading execution, negative. (Consolidated from internal Z-Spin Collaboration research notes up to v0.4.)

v1.1 (March 2026): Review-integration release, superseding v1.0. (1) Title lowered to Conditional-Candidate tier; (2) USp(10)/sp(5) notation convention fixed at first use and throughout; (3) condition accounting corrected to one structural (H-SPIN) + auxiliaries (H-BF$_-$), (H-QM); (4) B3-1 tier restated as CONDITIONAL-CANDIDATE; (5) new §8: stage-2 quaternionic route ($b = 2h^\vee(C_5) = 12$), executed under a second frozen registration, family CLOSED-NEGATIVE; Theorem F41.8 (25/2 No-Go) with Witten/APS/index-parity obstructions; (6) Terminal Stopping Rule registered in §11 (gates F-F41.9, F-F41.10); the proposed successor paper (F42) is retired — no successor B3 paper exists. Verification extended 25 → 31 checks, 3 → 4 guards; all stage-1 results preserved bit-identical.